



FACULTY	Agriculture, Engineering and Natural Sciences		
SCHOOL	Engineering and the Built Environment		
DEPARTMENT	Mechanical and Metallurgical Engineering		
SUBJECT	ENGINEERING MECHANICS II		
SUBJECT CODE	I3641NM		
DATE	Wednesday 14th May 2025		
DURATION	1 Hour 15 Minutes	MARKS	100

Continuous Assessment Test 1

Examiner: Prof. I. O. Ohijeagbon

This question paper consists of **2 pages** including this front page.

Instructions

1. Closed book examination
2. Read the questions carefully
3. This question paper consists of **Three** questions
4. Answer **any 2 Questions**
5. Each question carries **50 marks**
6. Neatly drawn sketches should be used where required
7. **Supplementary materials supplied:** None

Question One

(a) An investigation was carried out to study the rectilinear kinematic motion of a particle. Illustrate the possible analysis you would likely consider to describe the characteristics of the motion if the acceleration is constant.

$$v = ds/dt \quad ds = v dt \quad a ds = v dv$$

[16 Marks]

(b) During a test, a rocket travels upward at 75 m/s, and when it is at position $S_A = 40$ m from the ground its engine fails. While in motion the rocket is subjected to a constant downward acceleration of 9.81 m/s^2 due to gravity. By neglecting the effect of air resistance. Determine:

- (i) the maximum height S_B reached by the rocket,
- (ii) its speed just before it hits the ground, and
- (iii) the total time of flight of the rocket just before touching the ground.

[34 Marks]

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$$v = v_0 + at$$
$$80.06 = 0 + 9.81t$$

t

Question Two

(a) An investigation was carried out to study the curvilinear kinematic motion of a particle. Illustrate the possible analysis you would likely consider to describe the characteristics of the motion with respect to its rectangular components. [14 Marks]

14

(b) The velocity of a particle is given by $\mathbf{v} = \{16t^2\mathbf{i} - 4t^3\mathbf{j} + (5t + 2)\mathbf{k}\} \text{ m/s}$, where t is in seconds. If the particle is at the origin when $t = 0$. When $t = 2 \text{ s}$, determine:

- the magnitude of the particle's velocity,
- the magnitude of the particle's acceleration, and
- the x, y, z coordinate position of the particle at this instant.

36

[30 Marks]

Question Three

(a) The free-flight motion of a projectile is illustrated as shown in Figure Q. 3 (a), launched at point (x_0, y_0) , with an initial velocity of $\mathbf{v}_0$, having components $(v_0)_x$ and $(v_0)_y$. Starting from first principles, conduct the kinematic analysis of a projectile for:

- Horizontal motion
- Vertical motion

20

[14 Marks]

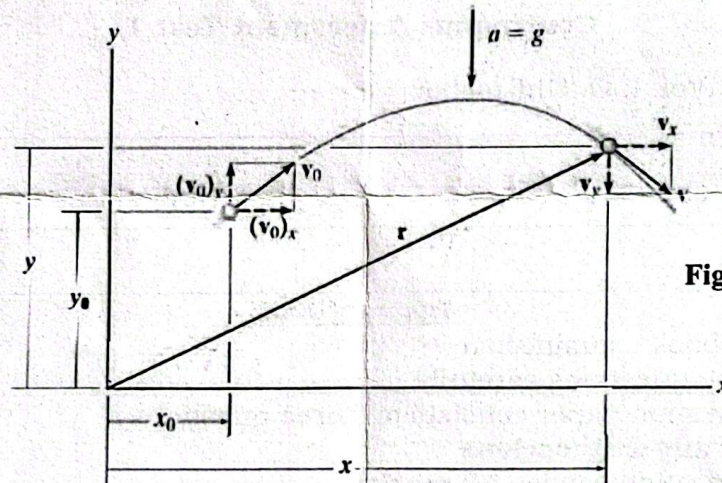


Figure Q. 3 (a)

(b) Pumping of water from a flooded area to a nearby canal over a landscape was required. Water is pumped through the hose into the air at a slope of 30° , from a height of 1 m from the ground. Projected water is suspended in the air before touching the ground after 1.5 s . Determine:

- the speed at which water exit the water hose,
- the horizontal distance water travels before striking the ground,
- the maximum height attained by projected water, and
- the components of the velocity when the water strikes the ground.

30

..... [36 Marks]